

Ethical Surveillance: Regulating for personal and surveillance data in COVID-19 control conditions

By Mark Findlay, Professorial Research Fellow, Director, Centre for AI and Data Governance, Singapore Management University

COVID-19 has prompted the roll out of a range of control measures and technologies adopted by authorities worldwide. As witnessed over the course of the past year, these technologies have been developed with the capacity to amass and share personal data for community control and citizen safety motivations. Nonetheless, by strengthening our existing surveillance infrastructures, they have also raised serious and as yet unanswered ethical and regulatory questions. This lecture reviews the ethical challenges—discrimination, lack of transparency, neglect of individual rights, and more—which have arisen from COVID-19 technologies and its resultant mass data accumulation. We offer a deeper look into the roll out of these control policies and technologies to regulate the movement of migrant workers in Singapore as a case study of these ethical challenges and trade-offs. We then suggest three normative foundations that need to lay the groundwork for regulatory exercises to address these ethical challenges and round off our discussion with regulatory strategies and policy recommendations.

Lecture Transcript

Mark Findlay

Welcome to the presentation on ethical surveillance regulating for personal and surveillance data in the COVID-19 control conditions. I'm Mark Findlay, the Director of the SMU Center for AI and Data Governance. With me are Nydia, Jane and Josephine, who are Research Associates at the Center, and who will be presenting various parts of the material for you today. The Center for AI and Data Governance, we are a center that was established a little over two years ago to examine regulation and governance questions. When the Singapore government was concerned about the fact that AI was being pushed out at a great rate, with a lot of promotion and sponsorship but regulation and governance questions perhaps had not been addressed as well as they might be, and therefore the brief looking at regulation and governance themes covering AI and big data is in the hands of our center. The project designed for the center is around three core areas - which the slide shows you - associated with those are some fairly predictable areas of concern like trustworthy AI, transforming the labor force, autonomous vehicles, these themes. But we also have a series of cross sector projects which control and connect the various individual project heads. And these look at things like questions of the apportioning of responsible ethics across the AI ecosystem. And we have a range of other regional interests that deal with connecting research centers into our common design. We have also been involved in a variety of areas of research that are COVID connected. Just very briefly, the rollout of code control measures and strategies has been largely a jurisdictionally based initiative and we'll be talking about the situation in Singapore today. But there are parallels that are easy to be drawn. And those parallels hopefully, you can see for your own jurisdiction with the challenges that arise. We have been looking at a range of issues that are associated with COVID control, and primarily the influence of surveillance on personal data. The lecture outline has three parts, control technologies, mass data accumulation, and challenges which look at the foundations for regulatory need and necessity, regulating for ethical outcomes, which is a process whereby we have a variety of different ethical concepts and strategies in place, and then some

AI ETHICS

recommendations in relation to what the future holds in a surveillance society. I'll pass you on now to Jane. Thank you.

Jane Loo 02:55

Thank you Mark for the introduction. So just to give a brief lecture outline, our lecture will follow this order. For the first part, we will cover the challenges that arise from these COVID control technologies and its mass data accumulation, we will put forward some recommendations outlining the ways in which these challenges can be addressed. We will then examine some of the surveillance strategies adopted in Singapore and how it has raised some challenges for the migrant worker population. Next, we will propose how to regulate for more ethical outcomes. For the final part of our lecture, we will draw attention to some of the emerging issues that surround post pandemic surveillance. These are the challenges that we will discuss in the slides to follow, we will first describe how COVID surveillance technologies have brought about each individual challenge. And then we will propose how to address these challenges in order to promote ethical surveillance. This challenge that we should concern ourselves with is the problem of discrimination. Discrimination can arise on at least three levels here. Firstly, because of the lack of early risk prediction, states end up using more invasive control measures on particular groups when less invasive measures become unavailable. Groups can also be discriminated against because the adopted tech itself is discriminatory in nature and in implementation. For example, when only a certain group is mandated to download a contact tracing app, but it remains optional for the wider public. Discrimination can also arise when data collected or harvested is used to adversely impact on a particular population. Promoting ethical surveillance then involves preventing discrimination and discriminatory outcomes. To do so we propose that governments place more emphasis on the need to protect the vulnerable and include anti-discrimination principles in the legal framework that regulate infectious disease control. Another ethical challenge that arises out of the use of COVID tech and its amassed data is when such data is used outside of the purpose of crisis containment, it threatens the right to human dignity. Today, it is understood that data can be used to review virus hotspots and potential exposure. However, it is also the case that such datasets can threaten individual dignity. This was the case in South Korea when a link was drawn between the virus and the gay nightclubs. As a result, LGBTQ individuals face heightened discrimination and harassment from their community. In rolling out these technologies, promoters must continue to safeguard and respect the right to human dignity. On the point of transparency, many promoters of these surveillance technologies do not review how the collected data is stored, utilized and repurposed. This lack of information interferes with the public trust. For example, in China, it was revealed that its Alipay health code app, which supports quarantine decisions, shares information with the police. This was not initially made known to the public. To promote ethical surveillance, the relevant authorities should aim to publicly enunciate the reasons behind any limitation of liberties and ensure that information delivered to the public is clear and accessible. Related to the above point on discrimination, COVID surveillance tech and its amassed data could produce biased pandemic containment efforts based on nationality, religion and race. Making public health decisions based on such datasets could leave out entire populations or misrepresent others. To promote ethical surveillance authorities, need to actively avoid biases in data collection and processing. Data collected and processed must be genuine and emphasis should be placed on securing data integrity. It is also important to monitor these databases for future bias generation. And finally, authorities should also be mindful of their own inherent biasness. The next challenge occurs. The next challenge concerns the lack of explain ability. It is observed that data subjects lack information about the legitimate purposes for many of the COVID surveillance measures that have been rolled out, including why these technologies are implemented and how the data is being used for pandemic containment purposes. This lack of explain ability is damaging to the overall public trust and confidence. Promoting ethical surveillance involves

AI ETHICS

ensuring data subjects comprehend the legitimate purpose for data harvesting and usage. All the risks and benefits associated with these surveillance technologies must be clearly enunciated to the public. On the next point, a lot of emphasis has been placed during this time on securing the public interest and health to the exclusion of fundamental individual rights, such as privacy and liberty. Responsible use of surveillance should factor in the principle of proportionality and respect for individual rights and liberties. On the next point, the crisis has created a climate of fear with elevated rates of anxiety among members of the public. The threat to one's personal health, as well as a fear of being accused of not being a good citizen, has prompted large scale willingness to compromise on one's individual rights. The lack of scrutiny in these measures is equally problematic. Ethical surveillance involves rejecting scaremongering tactics. To reduce public anxiety, policymakers should provide daily accurate reporting concerning details of infection rates to ensure that the public is kept up to date. Emphasis must also be placed on encouraging the voluntary adoption of any COVID surveillance tech. Apps should not be made mandatory to take advantage of the public sphere. Next, when public health insights are derived from aggregated location data, governments should be mindful of its potential for reidentification and its impact on personal privacy. For example, location data have the potential to review highly sensitive information that might expose an individual's personal life, subjecting them to discriminatory practices. The promotion of ethical surveillance calls for the implementation of better data protection practices to safeguard privacy. It is also important to ensure that data collected cannot be monetized or reverse engineered to track people for non-crisis purposes. Finally, one last challenge that arises out of the use of covert surveillance tech is the failure to recognize the importance of expiring and rolling back on the strategies post-pandemic. To promote ethical surveillance, it is essential that authorities see to the exploration and reduction of data once the purpose for its collection has been fulfilled. It is also important that users are given the option to review and delete their data at any time. With these ethical challenges in mind, it will now be useful to explore the Singapore migrant workers use case and see how exactly imposed COVID measures have impacted on this particular population. Just to provide a bit of context, Singapore has approximately 251,800 migrant workers and as of late July, they account for more than 90% of Singapore's infection rates. To curb the spread of the virus within their living spaces, the Singapore government implemented various specific COVID surveillance measures on these groups. This has raised several ethical challenges earlier discussed, including the problem of discrimination, the threat to the human dignity and rights, the lack of transparency and uncertainty as to the expiry of these measures. These challenges will be made obvious as we go through the measures themselves that have been rolled out. As mentioned earlier, several surveillance measures have been applied towards the migrant populations that are demographically specific to them. They include quarantine and isolation; the fact that the state has delegated its pandemic containment powers to private employers with vested economic and corporate interests; also how migrants are made to use several mandatory applications that are not mandated for the wider community. Firstly, because less intrusive options were closed off due to the lack of early prediction, all migrant dormitories in Singapore were placed under effective lockdown and isolation. Workers were therefore not permitted to leave the dormitories and were placed under surveillance for months. This measure is arguably discriminatory because it did not apply to wider society. The lockdown also brought out the worst of their living conditions since full time and shift workers were stuck in the same place at the same time. To date, workers are still not permitted to leave their dormitories without consent. They are only allowed out for work and are ferried to and from their workplaces and dormitories. Only recently were they granted permission to visit recreational facilities, but consent must still be sought. Notably, the restrictions placed on your freedom of movement are not time limited. Flowing from the measure of quarantine and isolation, the state delegated some of its pandemic containment powers to private employers with vested economic and corporate interests. Here are some of the powers that were delegated to private employers. Firstly, employers have to provide food to their migrant workforce. The power imbalance as between employer and migrants make the

AI ETHICS

structure of answerability very problematic because workers are not well positioned socially or economically to make such complaints. Secondly, the state instructed employers to ensure that migrant workers adhere to safe distancing measures. However, there was a lack of oversight by the state over the choice of what measures were allowed to be employed, and this discretion left employers' powers open to abuse. So, in one incident, to ensure safe distancing, a dormitory operator locked several migrants up in the dormitory bedrooms from the outside. Finally, amendments were made to the law asking employers to monitor the movement of their migrant workforce. As a result of the amendments, workers are forbidden to leave the accommodation without their employers' consent and are only allowed out in emergencies. The regulation itself lacked clear objective criteria to evaluate what an emergency should entail the above powers granted to private employers pose an obvious challenge to migrants' right to human dignity and other fundamental rights. To cut the spread of the virus the authorities have made the use of several apps' compulsory for the migrant population in Singapore.

TraceTogether is a contact tracing app utilizing Bluetooth technology. Workers use SG work cards to check for a green status on their access code in order to be granted permission to attend work. What is problematic here is that the workers ability to leave his dormitory for work and to be granted a green status would ultimately depend on the initiative of his employer. Therefore, if an employer fails to send his worker for a mandatory health check, the worker will not be permitted to leave his dormitory. For FWMOMCare, workers have to use a feature in the app to scan a QR code on the door to report the location at the start and end of work. The app uses GPS tracking. Workers are placed under heavy surveillance and they are tracked right down to whether they are in their rooms or not. In light of the ethical challenges earlier discussed, relevant questions must be asked here. Where is all this data being stored? Is data collected sufficiently protected? Is such extended surveillance truly necessary? Has the state sufficiently considered the principle of proportionality and are individual rights still respected? Finally, and most recently, the state started distributing blue pass tokens to migrant workers. The token has to be worn on your wrist. And this is arguably problematic because in reality such tagging may foster an environment of discrimination and stigma against individuals who are made to wear the wristband. The authorities claim that the tokens will help to identify and isolate the close context of infected persons. But no further information has been provided on the utilization, security and storage of the data collected on these devices. There is an overall lack of explainability and transparency as to how this measure will operate. In conclusion, by exploring some of these surveillance measures that have been ruled out by the Singapore government on the migrant camp, we see how several ethical challenges have arisen. They include the problem of discrimination, the threat to human dignity, lack of explainability and transparency and the uncertainty surrounding the expiration date and phasing out of these surveillance measures. In the slides to follow, Nydia will forward some recommendations outlining the ways in which surveillance tech and its amassed data can be better regulated. Thank you.

Nydia Remolina 14:56

Thank you so much, Jane. Now that Jane has provided the overview of the ethical challenges and a zoom into the case study of Singapore and what has happened with the COVID-19 control situation, we will now offer some suggestions regarding effective and inclusive regulatory responses when faced with extended surveillance, track and tracing, public private provider data sharing, and any breakdown in personal data firewalls, and we think these suggestions that we will offer to the public and to policymakers will be able to build and to help and contribute to promote the ethical surveillance that we all want. In approaching any regulatory enterprise, there are four fundamental features influencing the ultimate regulatory chose, choice and direction. And those questions are why, when, where, what and who should regulate and what to regulate? Where are the answers of these questions in order to start providing some solutions and recommendations for the regulatory model for ethical surveillance, or at least to contribute to the path towards ethical surveillance? First, let's talk about the why. Why do we need to

AI ETHICS

regulate mass data use in a controlled situation, especially in surveillance situations? The simple answer is that because control technology is employed for surveillance to store or disseminate personal data, then this should not proceed without responsible governance. That's logic. That's very intuitive for everyone. But the matter is not that simple. In practice, surveillance initiatives and any personal claims over data are always contextual, control situations, especially for regulators imply balancing objectives and challenges because they have to balance this public and the public interest with individual rights. So, there is this balance exercise is part of the answer that we need to provide when we ask ourselves and regulators ask why do we need to regulate? Does the why question becomes difficult to isolate from the consent, compliance, goodwill, or even reluctant acquisitions of the data subject? When is the second question? When do we need to regulate, especially when we're talking about situations as such as COVID-19 when an emergency is happening, and regulators and the government and the states need to put into practice and implement a surveillance initiative, how to build or contribute to build up that an ethical surveillance in those emergency situations? The COVID-19 situation shows that when the emergency conditions are sufficiently relieved, to return to considerations of conventional personal data protection, when to intervene, maybe more a political and economic rather than a health, science or technical determination. To avoid inconsequential deliberations over when is it safe enough to be concerned about personal data use, regulators can suggest it is more productive, to get protections in place, as we roll out and apply in process technologies. There is no crisis to great or no personal data too insignificant to obey the need for regulatory oversight. So when is it clear? We are affecting these rights and the use of personal data? Where? Where do we need to regulate? Again, this is this could be answered simply, wherever the data is produced, stored, accessed and used. Yet in the spirit that data has value for those on whose behalf we regulate regulatory activity, its location and reach will depend on how much the regulatory recipient wants something to be done. The value of personal privacy is militated by access to private space and familiarity with rights discourse. This is linked to what Jane actually was mentioning with the migrant work situation. Along with that Jane mentioned previously, the discriminatory resonance and the discourse for migrant workers confining hostels, prisoners and mental health patients, patients in secure facilities, residents in aged care institutions that we've witnessed in other countries, the poor in the slums, and people living on the streets should not justify regulatory location only where personal data and individual liberties are actionable. These should be broader and apply to everyone, every individual. What - what to regulate regulatory techniques. range across the continuum of command and control to the least intrusive compliance formats where any regulatory initiative sits will depend on the urgency cooperation and competing regulatory demands so against we link here this day with the balance of different interests, the public interest, or what the state calls the public interest, and individual interests and rights. Who - who should be regulating and who should be building the model for ethical surveillance? There is a need to identify conundrums that attach to attribution and risk distribution of responsibility. Specifically, when we're talking about the use of technologies such as AI mass data use, and data driven surveillance, we have seen that governments are partnering with private institutions and the big private sector. So distribution of responsibility becomes a very important question to answer. Ultimately, depending on who is that that advocates and promotes and administers control and surveillance technologies, then the responsibility to build strategies to avoid harm and misuse rests first with them. And in the next slide, we will address the general regulatory fundamentals of our regulatory model to make the choice and extrapolate the potentials of our regulatory strategy for ethical surveillance more focused, accessible and relevant to an audience with different views on regulatory need. The strategy is framed around three topologies of challenge to personal data as Jane was mentioning before. Individual liberty, integrity, authority, legitimacy and accountability and good governance and data justice in higher order discharging intense not to exacerbate negative consequences already featuring and emerging from control approaches. There are three encompassing normative foundations for the regulatory exercise that we are proposing that is obviously applicable to COVID-19 situations and pandemic

AI ETHICS

control type of surveillance, but it can be extrapolated, as I mentioned, to other type of surveillance towards building ethical surveillance in general. First, lessen and avoid discrimination. Regulatory regulation cannot cure all the structural inequalities prevailing around surveillance technologies and data use. That's pretty clear. But it can be mindful of these problems. Bias prevent both the data usage and its regulation fueling prevailing or emerging discrimination. Secondly, recognize and comply with established principles of ethical and data use, big data use and principle design that is advocated for AI. Paramount among these principles for our purposes are human dignity and solidarity when directed to individual liberty and integrity, transparency and explain ability when directed to authority, legitimacy and accountability, and fairness and harm avoidance when directed to good governance and data justice. And thirdly, promote citizen inclusion. While Protected Health and Safety controls tend to be paternalistic, they will then matter how well intentioned for the most supportive uptake require the brightest engagement across cohesive, simple and satisfactory opportunities for conflict resolution. There is going to be conflict when applying surveillance initiatives. So this is important. It is not enough for the state or the private sector data repositories to ask for compliance and unquestioned trust when many of the risks associated with surveillance and data usage are not candidly revealed and openly negotiated with civil society. So the promotion of status and inclusion is a big part of our regulatory proposal. In the next slide, we'll get into the regulatory risk big policy recommendations. As Jane explained the challenges posed by any ongoing application of interests of data harvesting technologies, created or augmented during crisis conditions and lax data sharing limitations enabling mass data applications for similar control justifications, both very grave ramifications for personal data integrity, and then betting off and this powering surveillance societies. Therefore, vigorous and powerful regulatory infrastructure and process need implementation as matters of urgency, specifically in emergency situation, when surveillance initiatives are being placed. There regulatory options set out here in this presentation are designed to be thematic and not prospective, we are mindful that there are going to be differences jurisdictions from to jurisdiction and across different regulatory challenges. In brief summary, it is proposed that this regulatory cornerstone should be created through three different proposals. First, an independent agency for researching potential personal data challenges transition out of the crisis or motives that leave that lead to surveillance, and have carriage of research and potential personal data challenges transition out of the crisis. It will have a public education consultation and complaints function. And in addition, it would act as a personal data access arbitrator. This is important for us in our proposal, because some people might think that the regulator the data protection regulator in particular, might or may comply with these same functions or may perform these same activities. But in reality, and specifically, when implemented some surveillance initiatives, we have seen that in some countries, the data protection framework is not even applicable to those type of initiatives when they are run by governments. So that's why we are talking about the importance of an independent agency for their research in their potential implications and to provide with this information to the public. Second part of their regulatory model is the enforced server relation units. Tasked with the responsible operation and eventual commissioning of surveillance technologies and their data repositories on a technology specific focus, this unit would determine compliance guidelines in consultation with the independent agency, we were talking about public and private stakeholders and civil society. And the third part of the regulatory model focus is focusing on civil society empowered initiatives. What are civil society empowerment initiatives? During the COVID-19 crisis, many countries and communities have seen the emergence of organized and informal community endeavors designed to assist a number [...] risk safety control message. This volunteer power base could be enhanced and institutionalized to assist and build ethical surveillance. We do think that civil society empowerment is very important. And it's important to institutionalize those types of interaction between civil society and surveillance initiatives in order to work towards ethical surveillance and to somehow work on the ethical challenges that Jane was talking about previously. However, in the next slide, we are mindful that they there may be two initial

AI ETHICS

reservations raised against the proposals that we just mentioned. One would be cost and complexity and the other one would be over reliance on state sponsorship. Responding to the cost and complexity concern, which no doubt locating why we prefer the establishment of a purpose design authority, there is nothing arguing against this location with a permanent and more generalized data protection administration. As I was mentioning before, an approach like this would protect against costly duplication and unnecessary overlap. In administrative capacity and operational infrastructure, if we do have an independent agency in charge of this matters. Over reliance on state sponsorship would be the other critique that we found around this proposal. And what we have to say here is that the proposals, and particularly the two last are self-regulation proposals in primary operation. Further, each of these three proposed tools appear beneath the earlier mentioned regulatory attribution of first resort. Those who are promoting the surveillance have initial responsibility to ensure that automatically produced personal data and sufficiently protected within the operation of the technology and data use, as is commonly understanding and enforced self-regulation models. Most data use challenges will be met at the lowest level of the regulatory pyramid. And this will be no exception, assume the promoters of the surveillance are acting in the public interest at large. And finally, we do think that this proposal and this regulatory model that aims to contribute good ethical surveillance is is going to be attractive and important for private and public sector because this helps them to build long term trust and best practice reputation, proof of responsible regulatory commitment, ethical inscription, and a desire to show the world that universal rights and safeguards do not have to join the scale of human lives lost as the critical measure of control competence. Now that we have covered what we consider can help to ensure ethical surveillance. We would like to wrap up the session with some final remarks. And my colleague, Josephine will be conducting this part of the session. So Jo, the floor is yours.

Josephine Seah 30:42

Thank you, Nydia. So to recap, in this lecture, we've really focused on the current experiences of digital surveillance systems developed to address COVID-19 in order to ground our discussion of the challenges of ethical surveillance. So Jane laid out these issues before exploring how they've played out in the experiences of migrant workers here in Singapore. And Nydia then delved into regulatory fundamentals and proposed strategies and policy recommendations that might allow us to address some of these ethical challenges in the context of regulating data specifically. We would like to close this lecture by then recontextualizing this discussion of COVID-19 surveillance, and our suggested policy recommendations within the wider scope of surveillance studies, and the concept of ethical surveillance itself. And we offer the following three routes of where further research is needed as we move out from this first year of COVID-19. So firstly, in this lecture, we focused on how to address some of these ethical challenges to data regulation. But given both, given the role that algorithms are increasingly playing in both the study of COVID to for example, epidemiological studies, and beyond, such as the use of algorithms for risk calculation and risk scoring, more work might be done to apply these same normative regulatory fundamentals towards asking what forms of algorithmic governance we find acceptable in our societies. How might we explore empirically the ways in which models of enforced self regulation can work? Secondly, as other scholars have pointed out, many of these digital surveillance systems potentially have very long lasting consequences for years to come. And these consequences are, as we have tried to touch on in our lecture, playing out differently for different groups in society, for disadvantaged and marginalized communities, not only are they encountering more surveillance, they are also encountering more invasive forms of surveillance. Given this, then our research questions and its implications are more fraught. How do we ensure that surveillance strategies, whether this takes the form of data collection, or even the automation of welfare and social services are ethical and that they remain ethical in the long run for these groups? Certainly, given the expansion of both public and private surveillance systems through the pandemic, we might also want to ask ourselves how to study the

AI ETHICS

inevitable surveillance creep, where methods of surveillance which have previously been unacceptable or critiqued before COVID-19 have now come to be quite normalized. So for example, some scholars have pointed out that the pandemic has really short circuited critiques of the surveillance state. Given that it was these very techniques of surveillance dependent control strategies which aided the containment of COVID-19 spread. And so it might be worth asking how contingent events of COVID-19 and its control strategies have really changed our understanding of the concept of ethical surveillance itself. And this has ramifications for the kinds of AI and ML applications which we end up supporting, building and deploying and of course, COVID new novel. So thank you very much for spending your time with us today. And we look forward and welcome your questions, comments, and your feedback.